Pai Theory Assignment#2

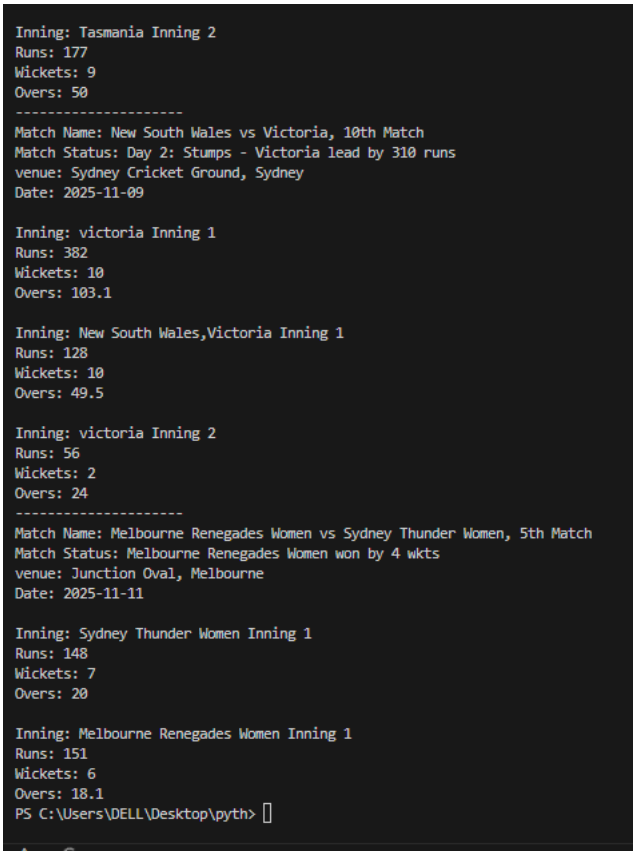
Arshia Azeem 24k-0012 BSAI-3A

Task1:

```
import requests
with open("api.txt",'r') as file:
    apiUrl=file.read()
    try:
        response=requests.get(apiUrl)
        if response.status_code==200:
            data=response.json()
            if data.get("status")=="success" and "data" in data and data["data"]:
                matchlist=data['data']
                print("API Status:")
                print(data['info'])
                for match in matchlist:
                    print("----------------------")
                    print(f'Match Name: {match['name']}')
                    print(f'Match Status: {match['status']}')
                    print(f'venue: {match['venue']}')
                    print(f'Date: {match['date']}')
                    if 'score' in match:
                        for score in match['score']:
                            print()
                            print(f'Inning: {score['inning']}')
                            print(f'Runs: {score['r']}')
```

```
                print(f'Wickets: {score['w']}')
                print(f'Overs: {score['o']}')
        else:
            errorMessage=data.get("reason","No live matches found or an unknown error occurred.")
            print(f"Could not retrieve matches:{errorMessage}")

    else:
        print(f"Error: Failed to fetch data. Status Code:{response.status_code}")
        print("Response:",response.text)
  except requests.exceptions.RequestException as e:
    print(f"A network error occured:{e}")
```

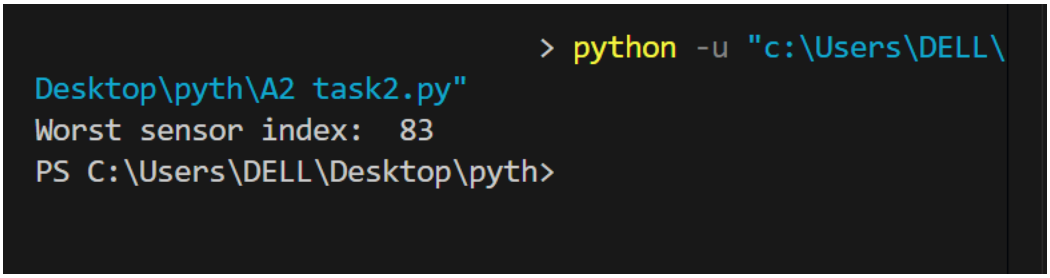
```
                        > python -u "c:\Users\DELL\Desktop\pyth\A2 task1.py"
API Status:
{'hitsToday': 9, 'hitsUsed': 1, 'hitsLimit': 100, 'credits': 0, 'server': 1, 'offsetRows': 0, 'totalRows': 6, 'queryTime': 24.731, 's': 0, 'cache': 1}
---------------------
Match Name: Bangladesh vs Ireland, 1st Test
Match Status: Day 1: Stumps - Ireland opt to bat
venue: Sylhet International Cricket Stadium, Sylhet
Date: 2025-11-11

Inning: ireland Inning 1
Runs: 270
Wickets: 8
Overs: 90
---------------------
Match Name: Pakistan vs Sri Lanka, 1st ODI
Match Status: Pakistan won by 6 runs
venue: Rawalpindi Cricket Stadium, Rawalpindi
Date: 2025-11-11

Inning: Pakistan Inning 1
Runs: 299
Wickets: 5
Overs: 50

Inning: Sri Lanka Inning 1
Runs: 293
Wickets: 9
Overs: 50
---------------------
Match Name: Western Australia vs Queensland, 12th Match
Match Status: Day 1: Stumps - Western Australia opt to bowl
venue: W.A.C.A. Ground, Perth
Date: 2025-11-11

Inning: Queensland Inning 1
Runs: 323
Wickets: 6
Overs: 91
---------------------
Match Name: Tasmania vs South Australia, 11th Match
Match Status: Day 2: Stumps - Tasmania lead by 209 runs
venue: Bellerive Oval, Hobart
Date: 2025-11-09

Inning: Tasmania Inning 1
Runs: 209
Wickets: 10
Overs: 64.2

Inning: South Australia Inning 1
Runs: 177
Wickets: 10
Overs: 48
```

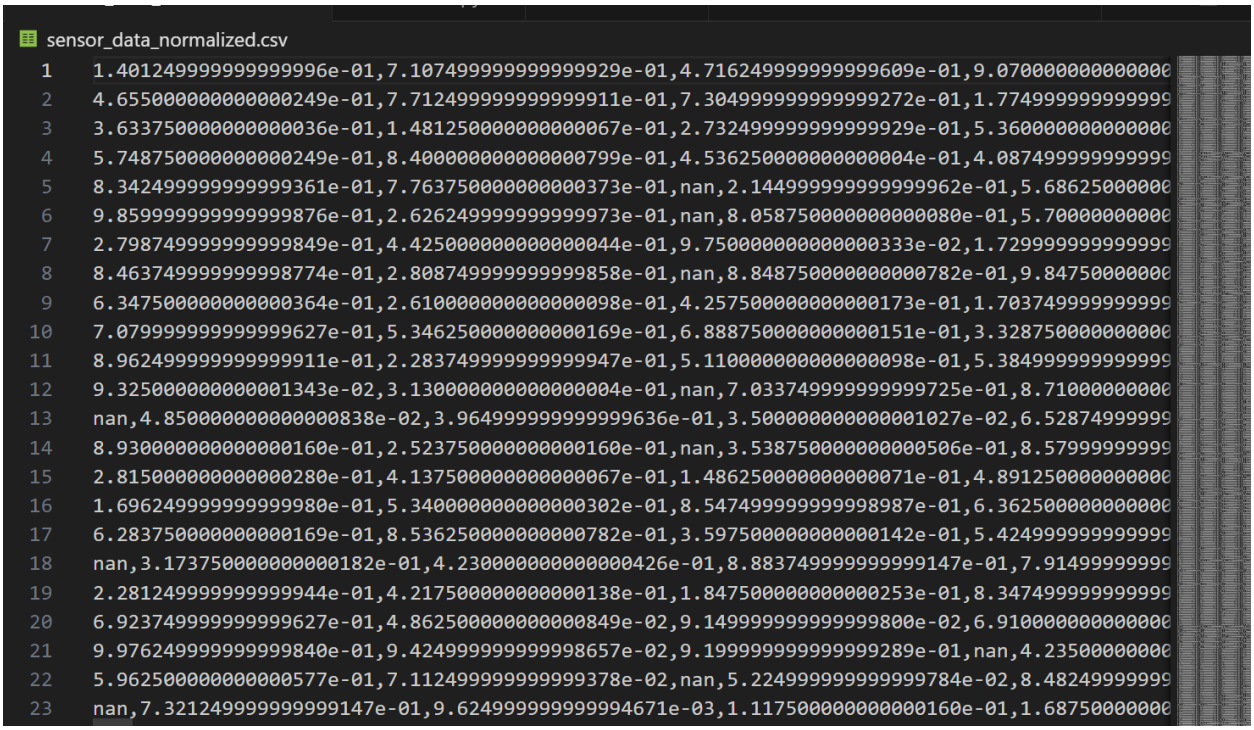


Task2:

```
import numpy as np
import csv
data=[]
with open("sensor_data.csv","r") as file:
    for line in file:
        row=line.strip().split(",")
        data.append(row)
arr=np.array(data,dtype=float)
arr[arr==-999]=np.nan
arr[(arr<0)|(arr>100)]=np.nan
```

```
sensor_mean=np.nanmean(arr,axis=0)

hour_median=np.nanmedian(arr,axis=1)

inv_count=np.isnan(arr).sum(axis=0)

worst_sensor=np.argmax(inv_count)

min_val=np.nanmin(arr)

max_val=np.nanmax(arr)

arr_norm=(arr-min_val)/(max_val-min_val)

np.savetxt("sensor_data_normalized.csv",arr_norm,delimiter=",")

print("Worst sensor index: ",worst_sensor)
```





Task3:

```
import pandas as pd
df=pd.read_csv("Titanic-Dataset.csv")
report=[]
for col in df.columns:
    dtype=df[col].dtype
    missing=df[col].isnull().sum()
    percentage=(missing/len(df))*100
    report.append(f"{col}: dtype={dtype}, missing={missing}({percentage:.2f}%)")
with open("inspect_report.txt","w") as f:
    f.write("\n".join(report))


df['Age'] = df.groupby(['Pclass','Sex'])['Age'].transform(lambda x: x.fillna(x.median()))
embarked_mode=df['Embarked'].mode()[0]
df['Embarked']=df['Embarked'].fillna(embarked_mode)
df=df.drop(columns=['Cabin'])
df['FamilySize']=df['SibSp']+df['Parch']
df['IsAlone']=(df['FamilySize']==0).astype(int)
df['Age']=df['Age'].astype('int64')
df.to_csv("titanic_cleaned.csv",index=False)
```

inspect_report.txt

```
PassengerId: dtype=int64, missing=0(0.00%)
Survived: dtype=int64, missing=0(0.00%)
Pclass: dtype=int64, missing=0(0.00%)
Name: dtype=object, missing=0(0.00%)
Sex: dtype=object, missing=0(0.00%)
Age: dtype=float64, missing=177(19.87%)
SibSp: dtype=int64, missing=0(0.00%)
Parch: dtype=int64, missing=0(0.00%)
Ticket: dtype=object, missing=0(0.00%)
Fare: dtype=float64, missing=0(0.00%)
Cabin: dtype=object, missing=687(77.10%)
Embarked: dtype=object, missing=2(0.22%)
```

Task4:

```
import pandas as pd


df=pd.read_csv("titanic_cleaned.csv")

fares=pd.read_csv("ticket_fares.csv")


df = df.merge(fares,on="Ticket",how="left")

df.columns = df.columns.str.strip()


if "Fare_y" in df.columns:
   df["Fare"] = df["Fare_y"]


bins=[0,12,19,59,120]; labels=['Child','Teen','Adult','Senior']

df['AgeGroup']=pd.cut(df['Age'],bins=bins,labels=labels, right=True)

survival_by_sex_age=df.pivot_table(values="Survived",index="Sex",
```

```
 columns="AgeGroup",aggfunc="mean")


print(survival_by_sex_age)

survival_by_sex_age.to_csv("survival_by_sex_age.csv")

class_survival=df.groupby("Pclass")["Survived"].mean()

print(class_survival)


class_survival.to_csv("class_survival.csv")

df["FareBin"]=pd.qcut(df["Fare"],q=4, labels=["Low","Medium","High","VeryHigh"])

fare_survival=df.groupby("FareBin")["Survived"].mean()

print(fare_survival)

fare_survival.to_csv("fare_survival.csv")


with open("report.txt","w") as f:

   f.write("Hypothesis 1: Women and children first.\n")

   f.write("Based on the survival rates grouped by sex and age group, women show a substantially higher survival rate than men across all age groups. Children, particularly female children, also show higher survival rates compared to adult males. These results support the hypothesis that 'women and children first' influenced survival outcomes.\n\n")

   f.write("Hypothesis 2: The wealthy had a higher survival rate.\n")

   f.write("Both methods used to approximate wealth—Passenger Class and Fare quantile bins—show that passengers in higher classes and those who paid higher fares had significantly higher survival rates. Therefore, the data supports the hypothesis that wealthier passengers were more likely to survive.\n")
```

```
blems (Ctrl+Shift+M)  LL\Desktop\pyth> python -u "c:\Users\DELL\Desktop\pyth\A2 task4.py"
c:\Users\DELL\Desktop\pyth\A2 task4.py:14: FutureWarning: The default value of observed=False is deprecate
d and will change to observed=True in a future version of pandas. Specify observed=False to silence this w
arning and retain the current behavior
  survival_by_sex_age=df.pivot_table(values="Survived",index="Sex",
AgeGroup    Child      Teen     Adult    Senior
Sex
female    0.38835  0.753247  0.699561  1.000000
male      0.32000  0.086957  0.202624  0.111111
Pclass
1    0.689055
2    0.480000
3    0.239057
Name: Survived, dtype: float64
c:\Users\DELL\Desktop\pyth\A2 task4.py:24: FutureWarning: The default of observed=False is deprecated and
will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or o
bserved=True to adopt the future default and silence this warning.
  fare_survival=df.groupby("FareBin")["Survived"].mean()
FareBin
Low         0.229426
Medium      0.441687
High        0.336449
VeryHigh    0.609418
Name: Survived, dtype: float64
PS C:\Users\DELL\Desktop\pyth>
```

report.txt

```
1   Hypothesis 1: Women and children first.
2   Based on the survival rates grouped by sex and age group, women show a
3   substantially higher survival rate than men across all age groups. Children,
4   particularly female children, also show higher survival rates compared to adult males.
5   These results support the hypothesis that 'women and children first' influenced survival
6   outcomes.
7
8   Hypothesis 2: The wealthy had a higher survival rate.
9   Both methods used to approximate wealth Passenger Class and Fare quantile
10  bins show that passengers in higher classes and those who paid higher fares
11  had significantly higher survival rates. Therefore, the data supports the hypothesis
12  that wealthier passengers were more likely to survive.
```